

Documentation to the L1outPRS program

Jurg Ott, Rockefeller University, New York

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This outline documents a simple implementation of a polygenic risk score (PRS) for GWAS data, that is, case-control data genotyped at a number of genetic variants, SNPs. It makes use of the `--score` function in *plink* [1-3], based on an allelic association test, `--assoc`. A PRS is computed for each case and control individual, with cases on average showing higher scores than controls. Classification (phenotype prediction) is performed by calling an individual a “case” if the individual’s score is above the 95th percentile of control scores, and otherwise is called a “control”. The raw classification is followed by cross-validation, that is, phenotype prediction is carried out for individuals not used for the classification procedure. The specific type of cross-validation implemented here is the Leave-one-out method [4], hence the program name, L1outPRS. The Pascal source code was compiled with the *fpc* package in Ubuntu (<https://www.freepascal.org/>).

The current version is dated 15 Jan 2025. Versions older than 10 Jan 2025 contained a bug that prevented execution for some jobs.

The L1outPRS program will compute PRSs for the best 5, 10, 20, ..., 50,000 variants (13 such sets of variants), where “best” refers to the ordering of variants selected by command line parameter #2 (see below). For each such selection, predicted phenotypes will be computed.

Once a prediction has been made for each individual, the following parameters are in general use:

Known phenotype	Predict “case”	Predict “control”
case	<i>a</i>	<i>b</i>
control	<i>c</i>	<i>d</i>

The letters, *a*, *b*, *c*, and *d*, indicate numbers of individuals, for example, *a* = number of cases predicted to be cases. Sensitivity, or power, of the test to identify cases is given by $a/(a + b)$; positive predictive value, PPV = $a/(a + c)$; negative predictive value, NPV = $d/(b + d)$; odds ratio, OR = $(a \times d)/(b \times c)$; prediction accuracy, ACC = $(a + d)/(a + b + c + d)$.

Included in this package is a case-control dataset, AMDHK, on wet age-related macular degeneration (AMD), collected in Hong Kong and published in 2006 [5]. The significant SNP in that publication is rs10490924 on chromosome 10.

The focus here is on prediction, not statistical significance of a case-control association test [6]. There is often no clear distinction between prediction and significance. Even in excellent textbooks [4], we find “Two useful summaries of predictive power are Sensitivity ..., Specificity”. However, sensitivity and specificity are properties of a statistical test and have more to do with statistical significance than prediction. For phenotype prediction, a stated aim of PRSs, a PRS is best based on highly predictive SNPs rather than significant SNPs [7].

L1outPRS program

This program creates a polygenic risk score (PRS) for single variants as implemented in *plink 1.9* with the allelic scoring function, `--score`. Because L1outPRS uses the Linux `sort` function (which is very different from `sort` in Windows), it runs properly only in Linux, for example, Ubuntu (Linux executable = L1outPRS). The *plink 1.9* executable must reside in the program path as `/usr/local/bin/plink19`. Download it from <https://www.cog-genomics.org/plink/>.

Command line parameters are as follows:

- 1) The *plink* fileset name, without “.map” and “.ped”, for example, AMDHKg. A minor allele frequency of at least 0.01 (preferably 0.05) should have been applied.
- 2) A number, 1 or 2, to indicate whether the variants are to be ordered by the *p*-value (by $1 - p$, actually) [1] or by the odds ratio [2].
- 3) A number, 1 or 2, to use prediction accuracy, ACC [1], or the positive predictive value, PPV [2], as the score variable.
- 4) *Optionally*: A number, *n*, indicating that calculations should only be carried out for the *n*-th best variant set rather than for all 13 variant sets.

An example of a command-line job submission is as follows:

```
jurg@Lap2023:~$ L1outPRS AMDHKg 2 1
```

Preferably start such a job in the foreground to see potential error messages and verify that everything is going well. If so, interrupt the job and submit it to run in the background:

```
jurg@Lap2023:~$ (L1outPRS AMDHKg 2 1 /dev/null)&
```

A report file, `L1outPRS.AMDHKg-2-1.rpt`, will be created in a directory above the running directory and will contain information on numbers of replicates processed. The output file will be named, for example, `L1outPRS.AMDHKg-21.out`. The four numbers, *a* ... *d*, in the output form a 2×2 table, which may be analyzed with the Fisher exact test [7]. The included utility program, *Fisher*, can carry out these calculations.

Currently the following maximum program constants apply:

- 10,000 individuals

References

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